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Community assembly of tropical Fagaceae-dominated forests in Thailand dates back at least to the Late Palaeogene

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The Late Oligocene to Early Miocene flora of the Ban Pa Kha Subbasin (Li Basin, northern Thailand) provides a record of montane dry tropical oak-pine forests. The rich ensemble of Fagaceae typical of these forests might have existed in the wider region of Southeast Asia since Eocene times and various fossil plant assemblages represented both lowland (Fagaceae, Dipterocarpaceae) and upland (Fagaceae, Pinaceae) tropical forests. These findings are in conflict with previous interpretations of vegetation development in northern Thailand, which stressed that stratigraphically older (possibly Late Oligocene) spore and pollen assemblages in northern Thailand were markedly different from the modern tropical flora and had a distinct northern temperate character. A major change in climate would have caused a dramatic shift to tropical conditions since the Mid-Miocene. Considering palaeobotanical data from adjacent regions in Southeast Asia, we suggest that differences in spore and pollen assemblages in intermontane basins in northern Thailand are more likely to represent different facies and lowland/upland settings. Assembly of these forest ecosystems, typically comprising *Quercus* sections *Cyclobalanopsis*, *Ilex* and *Quercus*, pollen of Castaneoideae with affinities to *Castanopsis* and *Lithocarpus*, and extinct fagaceous genera such as *Eotrigonobalanus*, most probably dates back to the Eocene. The absence of oaks of *Quercus* section *Cerris* in the spore and pollen assemblage of the Ban Pa Kha Subbasin, despite this group being part of the modern vegetation, might reflect the late arrival (secondary radiation) of this chiefly temperate group in tropical Southeast Asia.

ADDITIONAL KEYWORDS: Castaneoideae – fossils – Miocene – Oligocene – pollen – *Quercus*.

INTRODUCTION

Fagaceae (the oak family) comprise ten extant genera and > 1000 species of trees with a wide distribution across the Northern Hemisphere (Govaerts & Frodin, 1998; Phengklai, 2006; Denk & Bouchal, 2021; POWO, 2022) along with rare range extensions into the Southern Hemisphere at present (Govaerts & Frodin, 1998) and during the Palaeogene (Wilf *et al.*, 2019). The largest genus, *Quercus* L., occurs mainly in temperate to tropical regions of North America, Europe, Asia and parts of South America and Africa (Denk *et al.*, 2017; Hipp *et al.*, 2020). *Castanopsis* (D.Don) Spach and *Lithocarpus* Blume share their habitats and have a centre of diversity in the subtropical and tropical regions of southern China, south-eastern Asia and northern Australasia (Soepadmo, 1970, 1972; Phengklai, 2006, 2008; Strijk *et al.*, 2014; Strijk & Son, 2019; Zhou *et al.*, 2022). *Castanea* Mill. and *Fagus* L. are found in temperate to subtropical areas in North America, Europe and East Asia (Flora of North America Editorial Committee, 1997; Huang, Zhang &

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Bartholomew, 1999; Denk & Grimm, 2009; Zhou *et al.*, 2022). The small narrowly endemic genera *Chrysolepis* Hjelmq. and *Notholithocarpus* Manos, Cannon & S.H.Oh have a restricted distribution in western North America (Flora of North America Editorial Committee, 1997), whereas *Colombobalanus* Nixon & Crepet, *Trigonobalanus* Forman and *Formanodendron* Dixon & Crepet (subfamily Trigonobalanoideae) occur in Colombia, central and southern China, northern Thailand and central Malaysia (Nixon & Crepet, 1989; Phengklai, 2006).

On the basis of combined fossil and molecular analyses, the Fagaceae lineage probably diverged since the Mid-Upper Cretaceous, and the crown groups were established during the Palaeogene (Xing *et al.*, 2014; Grímsson *et al.*, 2016a; Hipp *et al.*, 2020; Zhou *et al.*, 2022). The oldest reliable fossil record of Fagaceae dates back to the Upper Cretaceous of North America (Herendeen, Crane & Drinnan, 1995; Grímsson *et al.*, 2016a). From the Early Cenozoic onwards, numerous fagaceous fossils appeared in North America (e.g. Manchester & Crane, 1983; Grímsson *et al.*, 2015; Bouchal, Zetter & Denk, 2016b), Europe (e.g. Krutzsch, 1959, 1967; Kvaček & Walther, 1989, 2010; Bouchal *et al.*, 2016a; Grímsson *et al.*, 2016b; Sadowski, Schmidt & Denk, 2020; Denk & Bouchal, 2021) and East Asia (e.g. Liu *et al.*, 2007, 2019; Li *et al.*, 2015a; Liu, Song & Jin, 2020; Naryshkina & Evstigneeva, 2020). Thus, based on the distribution of fossil records, the family had an extensive distribution in the Northern Hemisphere. The fossil records of this plant group are also key elements indicating biogeographical links between North America and Eurasia through the North Atlantic and Bering land bridges during the Cenozoic (e.g. Tanai & Uemura, 1994; Denk, Grímsson & Zetter, 2010, 2012; Grímsson *et al.*, 2016a; Sadowski, Hammel & Denk, 2018; Sadowski *et al.*, 2020).

Many previous pollen morphological studies using light, scanning electron and transmission electron microscopy have demonstrated the high diagnostic value and phylogenetic significance of pollen in Fagaceae (e.g. Crepet & Daghlian, 1980; Solomon, 1983a, 1983b; Denk & Grimm, 2009; Makino, Hayashi & Takahara, 2009; Bouchal *et al.*, 2014; Denk & Tekleva, 2014). In *Quercus*, pollen is diagnostic to the subgenus and sectional levels (Denk *et al.*, 2017), which has important implications for tracing the biogeographic histories of oak lineages in the fossil record (e.g. Grímsson *et al.*, 2015, 2016b; Bouchal *et al.*, 2017). Likewise, numerous palaeopalynological studies using a combined light microscopy and scanning electron microscopy approach dealt with dispersed pollen of Fagaceae from Cretaceous to Pleistocene strata across the Northern Hemisphere (Liu *et al.*, 2007; Hofmann, 2010; Bouchal *et al.*, 2014, 2016a, 2016b, 2017; Grímsson *et al.*, 2015, 2016a, 2016b; Li *et al.*, 2015b, Hofmann *et al.*, 2019; Sadowski *et al.*, 2020). In addition, several pollen studies included records of Fagaceae from Cenozoic strata of East Asia (Wang, 2006; Xu *et al.*, 2008; Qin *et al.*, 2011; Zhang *et al.*, 2012; Herman *et al.*, 2017).

The Ban Pa Kha Subbasin is located in the Li Basin, northern Thailand, where a rich plant fossil record representing pteridophyte, gymnosperm and angiosperm assemblages, derives from Oligocene to Miocene sedimentary rocks (Endo, 1963, 1967; Ratanasthien, 1984; Songtham *et al.*, 2003, 2005; Grote, 2005, 2015; Sawangchote, Grote & Dilcher, 2009, 2010; Grote & Srisuk, 2021). Fossil evidence from this Subbasin has been used to infer changes in palaeovegetation in northern Thailand caused by climatic shifts across the Oligocene–Miocene boundary and tectonic changes in Southeast Asia (Songtham *et al.*, 2003; Sepulchre *et al.*, 2010). From the Ban Pa Kha Subbasin, two fagaceous pollen taxa assigned to species of *Faguspollenites* G.V.Raatz and *Quercoidites* Potonié have previously been reported (Songtham *et al.*, 2003, 2005). Endo (1963, 1967) and Grote (2005) described six fossil species of *Quercus* section *Cyclobalanopsis* (Oerst.) Benth. & Hook.f. based on leaf remains from the Li Basin. A record of *Fagus feroniae* Unger [= *Alnus julianiformis* (Sternberg) Kvaček & Holý] reported by Endo (1967) must be excluded from Fagaceae. Various taxa of Fagaceae were also reported and partly figured with light microscopy micrographs, from Oligocene to late Mid-Miocene deposits in the wider area surrounding the Li Basin (Bande & Prakash, 1986; Sepulchre *et al.*, 2010; Sattraburut, Ratanasthien & Thasod, 2021).

In the present study, we investigated fossil pollen grains of Fagaceae from the Ban Pa Kha Subbasin, Li Basin, Lamphun Province, using a combined light microscopy (LM) and scanning electron microscopy (SEM) analysis to obtain the highest possible taxonomic resolution. The great diversity and systematic affinities of pollen of Fagaceae identified in our investigation along with a preliminary assessment of the entire spore and pollen assemblage from the Ban Pa Kha Subbasin formed the basis to infer the palaeoenvironment(s) for this subbasin during Late Oligocene to Early Miocene times. In addition, we discussed the timing of the assembly of modern forest types in Thailand.

MATERIAL AND METHODS

Geological setting of the Ban Pa Kha Subbasin

The Li Basin is a Cenozoic rift basin in the northern part of Thailand. Cenozoic sediments are exposed in the centre of the basin, whereas Palaeozoic rocks surround

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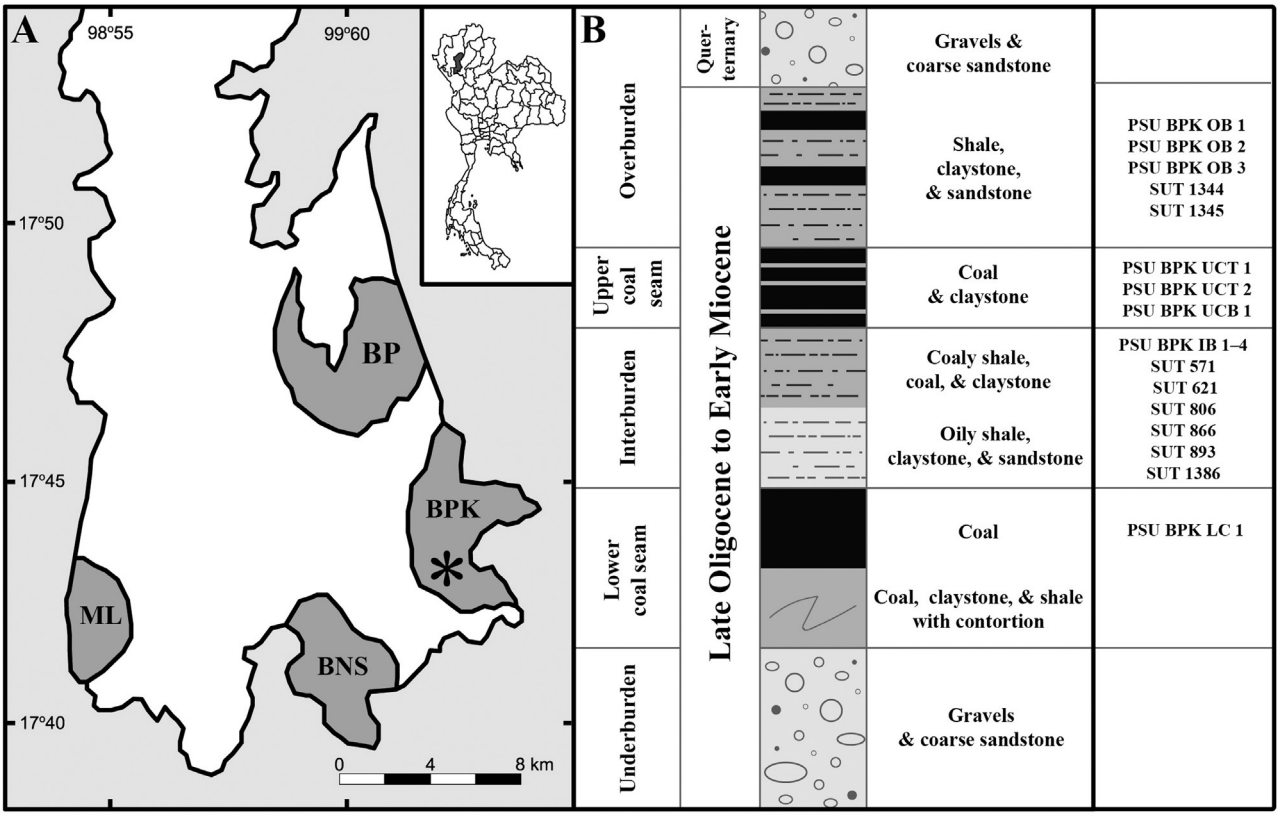

Figure 1. The geographical and geological setting of the Li Basin, northern Thailand. A, Map showing the geographical location of the Li Basin and its subbasins (BP = Ban Pu; BPK = Ban Pa Kha; BNS = Ban Na Sai; ML = Mae Long) (modified from Sawangchote *et al*., 2009, 2010). B, Stratigraphic chart illustrating sedimentary rocks of the Ban Pa Kha Subbasin (modified from Morley *et al*., 2001) and rock samples processed for the present study.

the basin (Morley & Racey, 2011). The depositional history of the basin was described by Nichols & Uttamo (2005). The Cenozoic sequence can be divided into two formations (Fm), the Oligocene to Early Miocene Li Fm comprising sandstone, mudstone and oil shale, and the Early to Mid-Miocene Mae Long Fm consisting of calcareous mudstone and lignite intercalations. The Li Basin comprises four main subbasins, Ban Pu and Ban Pa Kha subbasins in the east, Mae Long Subbasin in the south-west and Ban Na Sai Subbasin in the south (Fig. 1A) (Ratanasthien, 1990).

The sedimentary succession of the Ban Pa Kha Subbasin comprises five sedimentary rock layers *c.* 250 m thickness. According to Nichols & Uttamo (2005), the sediments accumulated in fluvial to lacustrine environments. The underburden layer, > 30 m thick, is composed of coarse sandstone with gravels. The lower coal seam, 15–20 m thick, consists mainly of coal, claystone and shale. The interburden layer, 10–15 m thick, is mostly composed of different types of shale. The upper unit of this layer is made up of coaly shale partly with claystone, whereas the lower unit is composed of oily shale with claystone and sandstone. These different layers are overlain by the 10–15 m thick upper coal seam, mainly composed of coal and partly claystone. The topmost overburden layer, 20–150 m thick, is composed of coarse sandstone with pebbles, claystone and shale (Fig. 1B; Morley, 1998; Morley *et al*, 2001; Sawangchote *et al.*, 2009, 2010; Morley & Racey, 2011). Numerous twigs, leaves, seeds and fruits of gymnosperms and angiosperms were found from the interburden layer (Grote, 2005; Sawangchote *et al.*, 2009, 2010), whereas dispersed plant spores and pollen were recovered from all sediment units (Songtham *et al.*, 2003, 2005).

Due to the lack of datable rocks or vertebrate remains, the small subbasins in the Li Basin are difficult to date. On the basis of the geological structure and tectonic evolution, the age of the Li rift basin was determined as the Late Oligocene to Early Miocene (Morley, 1998; Morley, Charusiri & Watkinson, 2011). Watanasak (1990) and Songtham *et al.* (2003, 2005) also suggested that within the Li Basin, basin filling of the Ban Pa Kha Subbasin took place from the Late Oligocene to the Early Miocene based on the presence of fossil pollen assigned to temperate elements. However, as

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pointed out by Morley *et al.* (2011), dating solely based on palynological data is difficult and a number of plant macrofossils from the Ban Pa Kha Subbasin could be affiliated with subtropical/tropical taxa (Grote & Srisuk, 2021). Nevertheless, for the present study we accept a Late Oligocene to Early Miocene age for the Ban Pa Kha Subbasin.

Sample preparation

Nineteen sedimentary samples from the Ban Pa Kha Subbasin were collected from the overburden, upper coal seam, interburden and lower coal seam (Fig. 1B). Each sediment sample was processed following the standard procedure by Halbritter *et al.* (2018). A combination of 37% hydrofluoric acid (HF) and 48% hydrochloric acid (HCl) was used to dissolve carbonate and silica. Acetolysis was used for removing organic materials other than sporomorphs. Fossil pollen grains from the residue were further examined with light (BX43, Olympus) and scanning electron (Jeol6400, FEI Quanta400 and SU3900, Hitachi) microscopy using the single grain technique (Zetter, 1989; Halbritter *et al.*, 2018). The different fagaceous pollen types were identified by comparison with published accounts on extant and fossil pollen taxa (e.g. Crepet & Daghlian, 1980; Liu *et al.*, 2007; Denk & Grimm, 2009; Hofmann, 2010; Bouchal *et al.*, 2014; Sadowski *et al.*, 2020). Chiefly, the pollen size, outline, exine thickness, apertural condition and exine ornamentation were examined. The terminology to describe pollen grains follows Punt *et al.* (2007), Denk & Grimm (2009) and Halbritter *et al.* (2018).

RESULTS

Fifteen fossil pollen taxa of Fagaceae from subfamilies Castaneoideae and Quercoideae were observed from the Ban Pa Kha Subbasin, Li Basin, with high percentages (Table 1).

Subfamily: not determined

Genus: Eotrigonobalanus Walther & Kvaček (extinct)

Eotrigonobalanus sp. 1 (Fig. 2A–C)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 29–30 µm long in LM, 28–29 µm in SEM, equatorial diameter 27–28 µm in LM, 26–27 µm in SEM; tricolporate, colpi long, pori elongated rectangular (lalongate); exine 1.1–1.2 µm thick (LM); tectate; sculpture scabrate in LM, microrugulate and perforate in SEM, microrugulae twisted and interwoven, microrugulae 0.6–0.9 µm long (SEM).

Remarks: Eotrigonobalanus is an extinct genus originally based on highly polymorphic leaves (Kvaček & Walther, 1989). Leaf remains have been described

Table 1. Percentages of fagaceous and pinaceous fossil pollen recovered from the Ban Pa Kha Subbasin

Stratigraphy of the Ban Pa Kha Sub-basin, Li Basin	Specimen number (processed rock samples)	*Eotrigonobalanus* & *Quercus* (%)	Castaneoideae (%)	Pinaceae (%)	Other pollen taxa (%)	Total pollen counts (grains)
Overburden	PSU BPK OB 1	70	0.4	0.4	29.2	506
	PSU BPK OB 2	75.9	0.4	0.6	23.1	540
	PSU BPK OB 3	88.3	0.4	0.6	10.7	504
	SUT 1344	85.8	0.4	4.9	8.9	527
	SUT 1345	87.9	0	3.4	8.7	552
Upper Coal Seam	PSU BPK UCT 1	24.1	0.8	0.8	74.3	510
	PSU BPK UCT 2	49.6	0.8	18.7	30.9	534
	PSU BPK UCB 1	41.7	1.4	1.7	55.2	511
Interburden	PSU BPK IB 1	76	0	8.8	15.2	500
	PSU BPK IB 2	90.8	0	4.2	5	520
	PSU BPK IB 3	81	0.4	11.7	6.9	531
	PSU BPK IB 4	83.5	0	9.5	7	599
	SUT 571	78.2	0.6	6.8	14.4	863
	SUT 621	76.7	0.8	5.7	16.8	541
	SUT 806	79.1	0.2	7.8	12.9	511
	SUT 866	79.7	0.6	8.6	11.1	620
	SUT 893	68.3	0.4	17	14.3	575
	SUT 1386	15.5	0.8	5.5	78.2	523
Lower Coal Seam	PSU BPK LC 1	0	0	0	0	0

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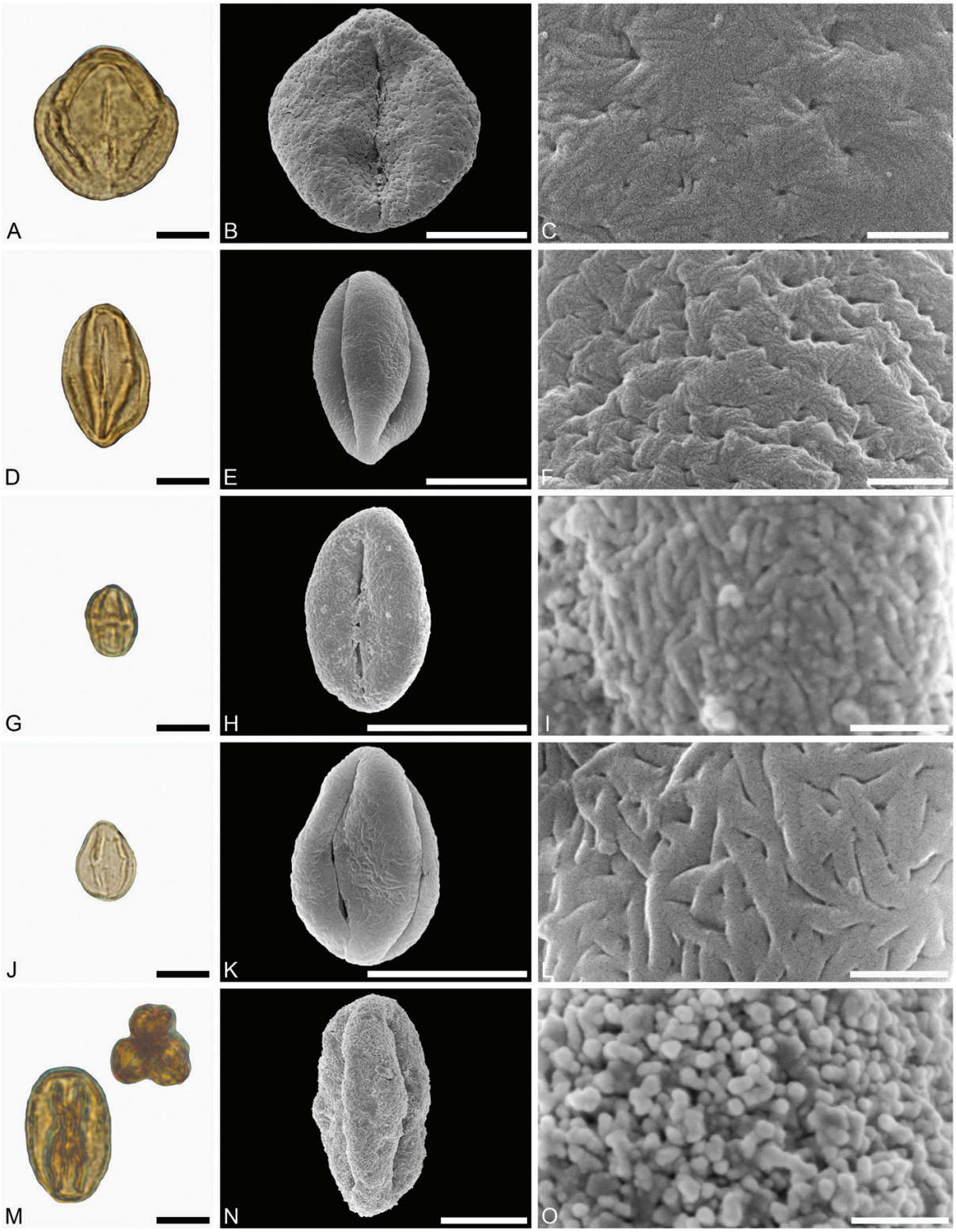

Figure 2. Fossil Fagaceae pollen from the Ban Pa Kha Subbasin. Light microscope (A, D, G, J and M) and scanning electron microscope (B, C, E, F, H, I, K, L, N and O) micrographs. A–C, *Eotrigonobalanus* sp. 1; D–F, *Eotrigonobalanus* sp. 2; G–I, Castaneoideae gen. & spec. indet. 1; J–L, Castaneoideae gen. & spec. indet. 2; M–O, *Quercus* section *Cyclobalanopsis* sp. 1. Scale bars: 10 µm (A, B, D, E, G, H, J, K, M and N) and 1 µm (C, F, I, L and O).

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from the Palaeocene (Danian) of Agatdalen, Nuussuaq Peninsula, Western Greenland (Grímsson *et al.*, 2016a), but are mostly known from the Mid-Eocene to Early Oligocene of Europe (Kvaček & Walther, 1989; Palamarev & Mai, 1998). In southern Europe, the leaf record seems to extend into the Miocene (Velitzelos, Bouchal & Denk, 2014). Fossil pollen showing comparable morphological features as the pollen described here (*Eotrigonobalanus* sp. 1 and sp. 2), has been found in clumps adhering to fossil leaves of *E. furcinervis* (Rossm.) Walther & Kvaček in Oligocene strata of Germany (Walther & Zetter, 1993), hence establishing its generic association. North American fossil pollen assigned to this genus has been described from the Upper Cretaceous (Early Campanian) of Wyoming, north-western USA; the Palaeocene (Danian) of Agatdalen, Nuussuaq Peninsula, western Greenland; the late Early Eocene (Ypresian) Princeton Chert, British Columbia, Canada and the Mid-Eocene (Late Lutetian to Early Bartonian) of Hareøen, western Greenland (Grímsson *et al.*, 2015, 2016a). Further, dispersed pollen from Oligocene strata of eastern North America were assigned to *Eotrigonobalanus* (Prader *et al.*, 2020). European *Eotrigonobalanus* pollen records include dispersed grains from the Palaeocene/Eocene of Salzburg, Austria (Hofmann, 2010; Hofmann, Mohamed & Egger, 2011), the Early Oligocene of Cospuden, Germany (Denk *et al.*, 2012) and the Mid-Miocene of Poland (Kohlmann-Adamska & Ziembińska-Tworzydło, 2000) and in situ pollen from anthers of staminate inflorescences preserved in Eocene amber from the Sambia Peninsula (Sadowski *et al.*, 2020). *Eotrigonobalanus* pollen has also been described from Early Miocene strata of Asia Minor (Denk, Güner & Bouchal, 2019), but to our knowledge the present record of *Eotrigonobalanus* sp.1 and sp. 2 is the first from Southeast Asia.

Eotrigonobalanus sp. 2 (Fig. 2D–F)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 27–28 µm long in LM, 24–25 µm in SEM, equatorial diameter 16–17 µm in LM, 14–15 µm in SEM; tricolporate, colpi long, pori elongate rectangular (lalongate); exine 0.7–0.8 µm thick (LM); tectate; sculpture scabrate in LM, microrugulate and perforate in SEM, microrugulae twisted and interwoven, microrugulae 0.3–0.8 µm long, microrugulae fused forming larger rope-like rugulae (SEM).

Remarks: Eotrigonobalanus sp. 2 differs from *Eotrigonobalanus* sp. 1 by the size of the microrugulae and the pollen wall thickness. The microrugulae are both narrower and shorter in *Eotrigonobalanus* sp. 2, and the wall is thinner. The *Eotrigonobalanus* sp. 2 type is more comparable to *Eotrigonobalanus* pollen from the Cenozoic of North America and Europe (see Remarks for *Eotrigonobalanus* sp. 1). Although we recognize two distinct (pollen) fossil species, it is possible that they reflect the morphological range of a single biological taxon. Previous studies on both fossil pollen (e.g. Grímsson *et al.*, 2015, 2016a) and extant pollen (e.g. Makino *et al.*, 2009) of Fagaceae have shown that several fagaceous taxa exhibit highly variable pollen grains considering their pollen size and particularly sculpture arrangement. In contrast, Sadowski *et al.* (2020) established different fossil species of *Eotrigonobalanus* from Eocene Baltic amber deposits based on distinct male flowers with in situ pollen. In this case, different fossil species produced nearly identical pollen morphologies.

Subfamily: Castaneoideae

Genus: not determined

Castaneoideae gen. & spec. indet. 1 (Fig. 2G–I)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 13–14 µm long in LM, 11–12 µm in SEM, equatorial diameter 9–10 µm in LM, 7–8 µm in SEM; tricolporate, colpi long, pori large and lalongate, nexine slightly thickened around pori (LM); exine 0.7–0.8 µm thick (LM); tectate; sculpture psilate to scabrate in LM, microrugulate, fossulate and perforate in SEM, microrugulae distinct, 0.5–2.0 µm long and 0.15–0.30 µm wide, microrugulae with a granulate suprasculpture (SEM).

Remarks: Pollen produced by all extant genera of subfamily Castaneoideae (*Castanea*, *Castanopsis*, *Chrysolepis*, *Lithocarpus* and *Notholithocarpus*) show similar morphology when studied with both LM and SEM (Crepet & Daghlian, 1980; Miyoshi, 1982; Manos, Cannon & Oh, 2008; Naryshkina & Evstigneeva, 2017). Although most species of Castaneoideae have small prolate pollen grains with microrugulate sculpture, the high relief and clear outline of microrugulae (observed with SEM) in extant *Castanopsis* and most *Lithocarpus* spp. are diagnostic compared to that of *Castanea*, *Chrysolepis* and *Notholithocarpus*, in which the microrugulae are more flattened and fused (e.g. Praglowski, 1982, 1984; Wang & Pu, 2004; Manos *et al.*, 2008). Therefore, it is likely that Castaneoideae gen. & spec. indet. 1 and 2 pollen is either from *Castanopsis* or *Lithocarpus*. Both genera are currently represented by tens of species in Thailand (Phengklai, 2008; Strijk *et al.*, 2014; Strijk & Son, 2019). Fossil

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Castaneoideae pollen grains are typical elements of numerous palynofloras from the Upper Cretaceous and Cenozoic of the Northern Hemisphere. According to summaries in Muller (1981) and Stuchlik *et al.* (2014), the earliest Castaneoideae-type pollen records are from the Santonian/Campanian of North America and Campanian of Europe. According to these occurrences, the parent plants producing this pollen type must have dispersed across the Northern Hemisphere before the Late Eocene. In North America, pollen of this lineage studied with combined LM and SEM has been recorded from the Upper Cretaceous (Early Campanian) of Wyoming, north-western USA; the Palaeocene (Danian) of Agatdalen, Nuussuaq Peninsula, western Greenland; the late Early Eocene (Ypresian) Princeton Chert, British Columbia, Canada (Grímsson *et al.*, 2016a); the Mid-Eocene (Late Lutetian to Early Bartonian) of Hareøen, western Greenland (Grímsson *et al.*, 2015) and the Late Eocene (Priabonian) of Florissant, Colorado, USA (Bouchal *et al.*, 2014). All European Cenozoic records before 2014 were summarized by Stuchlik *et al.* (2014) under the form-genus *Cupuliferoipollenites* Potonié. From the Russian Far East, Naryshkina & Evstigneeva (2020) described Early Eocene (Ypresian) pollen grains with a morphology highly similar to modern *Castanopsis*.

Castaneoideae gen. & spec. indet. 2 (Fig. 2J–L)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 15–16 µm long in LM, 13–14 µm in SEM, equatorial diameter 11–12 µm in LM, 9–10 µm in SEM; tricolporate, colpi long, pori small; exine 0.6–0.7 µm thick (LM); tectate; sculpture scabrate in LM, microrugulate, fossulate and perforate in SEM, microrugulae 0.5–2.0 µm long, 0.2–0.4 µm wide.

Remarks: Castaneoideae gen. & spec. indet. 2 differs from pollen of Castaneoideae gen. & spec. indet. 1 especially in the aperture region. The indet. 1 type has conspicuous endopori and the nexine is thickened around the pori. This is not observed in the indet. 2 pollen type. Also, the size of the microrugulae is smaller in the indet. 1 pollen and pollen grains have a granulate suprasculpture not noted for the indet. 2 type.

Subfamily: Quercoideae

Genus: Quercus L.

Remarks: Since the early 1980s, several publications emphasized the importance of pollen morphology (combined LM and SEM), and ultrastructure (TEM), for the circumscription of Fagaceae at the subfamily/genus levels (e.g. Crepet & Daghlian, 1980; Miyoshi, 1982; Wang & Pu, 2004) and especially for the infrageneric classification of *Quercus* (e.g. Solomon 1983a, 1983b; Denk & Grimm, 2009; Makino *et al.*, 2009; Denk & Tekleva, 2014). Research on extant *Quercus* pollen demonstrated that by using LM features (pollen size, wall thickness and aperture arrangement/peculiarities) and sculpture elements overserved with SEM, it is possible to discriminate major subgenera and sections of genus *Quercus* (e.g. Denk & Grimm, 2009). The only exception is that sections *Quercus* and *Lobatae* Loudon (red and white oaks) have several taxa that produce comparable pollen, and the small sections *Ponticae* Stefanoff., *Protobalanus* (Trelease) Schwarz and *Virentes* Loudon correspond largely with the pollen morphology of white oaks. In *Quercus* subgenus *Cerris* Oerst. (sections *Cerris* Dumort., *Cyclobalanopsis* and *Ilex* Loudon), pollen sculpture is highly useful for recognition of sections.

Subgenus: Cerris

Section: Cyclobalanopsis

Quercus section *Cyclobalanopsis* sp. 1 (Fig. 2M–O)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 26–27 µm long in LM, 25–26 µm in SEM, equatorial diameter 15–16 µm in LM, 13–14 µm in SEM; tricolporate, colpi long, pori small and indistinct; exine 0.6–0.7 µm thick (LM); tectate; sculpture scabrate in LM, nanorugulate to nanogemmate (rod-shaped) and perforate in SEM, nanorugulae/gemmae 0.1–0.2 µm wide, irregularly arranged, diverging and with a round protruding tip (SEM).

Remarks: The SEM-based pollen morphology of numerous Asian oaks belonging to section *Cyclobalanopsis* has been described/illustrated by Miyoshi (1982); Wang & Pu (2004); Fujiki & Ozawa (2007); Makino *et al.* (2009); Denk & Grimm (2009) and Miyoshi, Fujiki & Kimura (2011). Pollen in section *Cyclobalanopsis* has also been thoroughly studied with LM by Wang & Pu (2004), but see also Wang *et al.* (1995) and Miyoshi *et al.* (2011). Based on what has been published to date, it is clear that the SEM sculpture of the fossil *Quercus* section *Cyclobalanopsis* sp. 1 pollen type is unique and uncommon among modern taxa. This is expressed in the round protruding tips of the rod-shaped sculpture elements versus nanoechinate (pointed tips) in most extant species (e.g. Denk & Grimm, 2009). We are aware of only a single species, *Quercus championii* Benth. (see fig. 26 in Crepet & Daghlian, 1980), producing pollen with comparable sculpture elements. *Quercus championii*

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currently occurs at low- to mid-elevations (100–1700 m) in Taiwan and south-western to southern China (De Langhe, 2022). The fossil pollen record of *Quercus* section *Cyclobalanopsis* is meagre and only a few isolated grains investigated with combined LM and SEM have been assigned to this section. These include records from the Eocene of Hainan (Hofmann, 2010), the Miocene of Zhejiang (Liu *et al.*, 2007), China, and the Late Eocene of Florissant (Bouchal *et al.*, 2014), Colorado, USA and a putative record from the Mid-Miocene of Lavanttal, Austria (Grímsson *et al.*, 2016b). Still, none of these is comparable with the *Quercus* section *Cyclobalanopsis* sp. 1 pollen type described here.

Quercus section *Cyclobalanopsis* sp. 2 (Fig. 3A–C)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 19–20 µm long in LM, 18–19 µm in SEM, equatorial diameter 16–17 µm in LM, 14–15 µm in SEM; tricolporate, colpi long, pori small and indistinct; exine 0.7–0.8 µm thick (LM); tectate; sculpture scabrate in LM, micro- to nanorugulate (rod-shaped) and perforate in SEM, micro/nanorugulae indistinct, 0.4–0.8 µm long, 0.10–0.25 µm wide, irregularly arranged and clustered (SEM).

Remarks: *Quercus* section *Cyclobalanopsis* sp. 2 is similar to the previous type, but the sculpture elements are not as distinct or protruding. Whether these pollen types are representing the variation of a single biological taxon or actual species differences is not clear. It has been suggested by Denk & Grimm (2009) that secondarily accumulated sporopollenin can fill the 'space' between rugulae (rod-shaped elements) and cover their lower parts. Apparently, this can vary considerable between and within species (see fig. 4 in Denk & Grimm, 2009, and fig. 6 in Makino *et al.*, 2009). In any case, extant *Quercus gilva* Blume, a mid-elevation oak species currently occurring at 300–1500 m in central China and Japan (De Langhe, 2022) is known to produce pollen showing comparable sculpture (Makino *et al.*, 2009).

Quercus section *Cyclobalanopsis* sp. 3 (Fig. 3D–F)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 25–26 µm long in LM, 24–25 µm in SEM, equatorial diameter 20–21 µm in LM, 18–19 µm in SEM; tricolporate, colpi long, pori small and indistinct; exine 1.0–1.1 µm thick (LM); tectate; sculpture scabrate in LM, areolate to rugulate, fossulate and perforate in SEM, areolae/rugulae with nanorugulate (rod-shaped) to nanoechinate suprasculpture, nanorugulae and nanoechini irregularly distributed, distinct (SEM).

Remarks: Of the three *Cyclobalanopsis* pollen types presented herein, this is the most typical for the section. Most species of *Quercus* section *Cyclobalanopsis* produce pollen showing this diagnostic areolate/rugulate basal sculpture with numerous minute suprasculpture elements. For comparison, see for example the SEM pollen sculpture of *Quercus acuta* Thunb., *Quercus gilva* Blume, *Quercus glauca* Thunb., *Quercus hondae* Makino, *Quercus myrsinifolia* Blume, *Quercus salicina* Blume and *Quercus sessilifolia* Blume in Fujiki & Ozawa (2007), Denk & Grimm (2009); Makino *et al.* (2009) and Miyoshi *et al.* (2011).

Section: Ilex

Quercus section *Ilex* sp. 1 (Fig. 3G–I)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 23–24 µm long in LM, 23–24 µm in SEM, equatorial diameter 16–17 µm in LM, 15–16 µm in SEM; tricolporoidate, colpi long, pori small and indistinct; exine 0.6–0.7 µm thick (LM); tectate; sculpture scabrate in LM, micro- to nanorugulate (rod-shaped) and perforate in SEM, micro/nanorugulae 0.3–0.8 µm long, 0.1–0.2 µm wide, irregularly arranged, clustered (SEM).

Remarks: The pollen morphology (LM and SEM) and ultrastructure (TEM) of *Quercus* section *Ilex* has been studied in detail by Denk & Tekleva (2014), but see also Wang & Pu (2004); Denk & Grimm (2009) and Li *et al.* (2011). Oaks producing pollen grains that show similar sculpture as the fossil *Quercus* section *Ilex* sp. 1 type include Mediterranean and Asian taxa (De Langhe, 2022) such as *Quercus alnifolia* Poech, *Quercus aucheri* Jaub. & Spach, *Quercus baronii* Skan and *Quercus lanata* Sm. (see figs 7–10 in Denk & Tekleva, 2014). Fossil *Quercus* section *Ilex* type pollen studied with combined LM and SEM has been recorded from the Palaeocene/Eocene of Austria, the Eocene of Hainan, China (Hofmann, 2010), the late Early Eocene (Ypresian) Princeton Chert, British Columbia, Canada (Grímsson *et al.*, 2016a), the Early Oligocene of Cospuden, Saxony, Germany (Denk *et al.*, 2012), the Mid-Miocene of Lavanttal, Austria (Grímsson *et al.*, 2016b), the Mid-Miocene of Anatolia, Turkey (Bouchal *et al.*, 2016a, 2017; Bouchal, 2019), the Late Miocene of north-western Greece (Bouchal *et al.*, 2020), the Pleistocene of south-eastern Albania (Denk *et al.*, 2021b) and from Holocene sediments of the Sea of Japan (Tekleva, Naryshkina & Evstigneeva,

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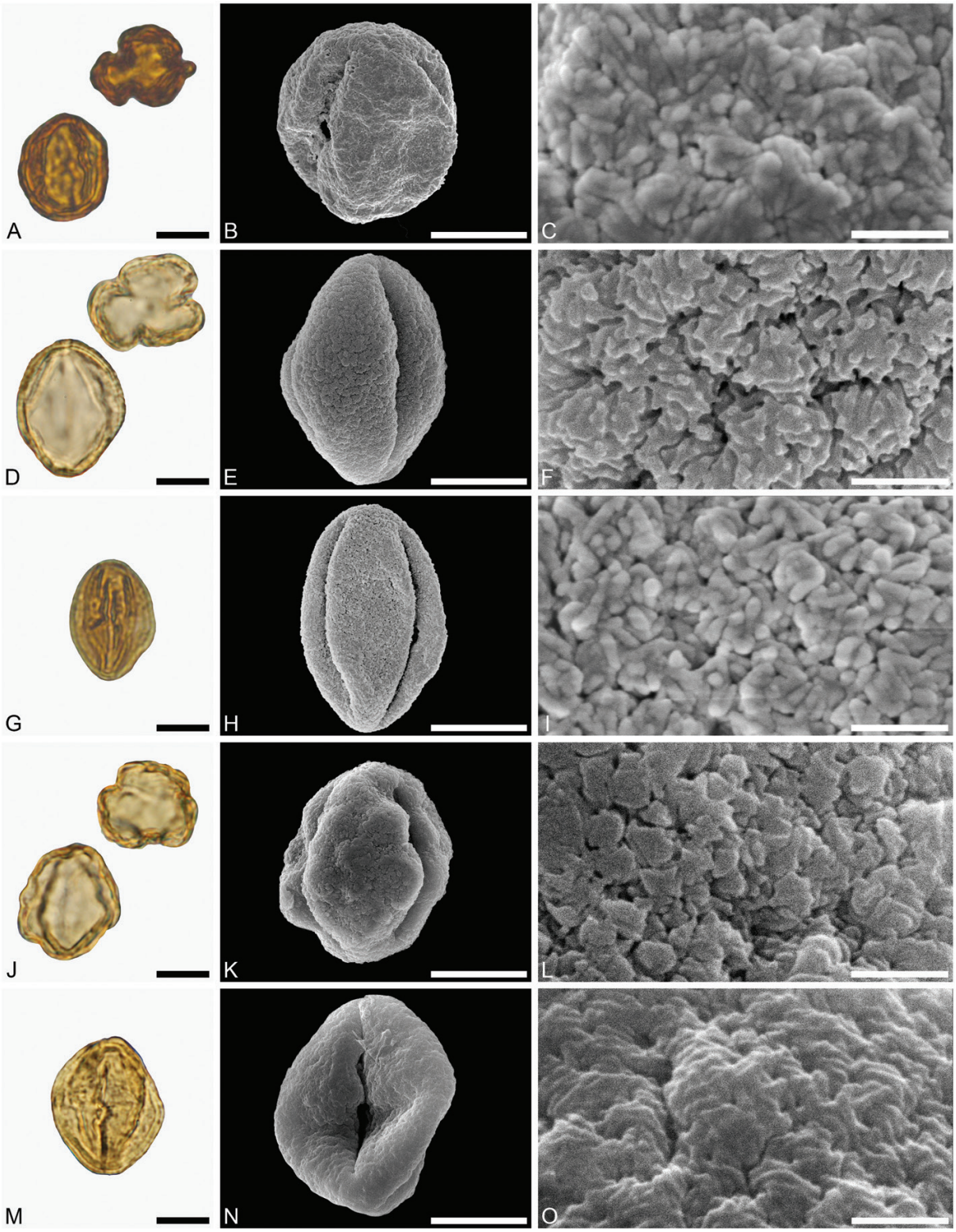

Figure 3. Fossil Fagaceae pollen from the Ban Pa Kha Subbasin. Light microscope (A, D, G, J and M) and scanning electron microscope (B, C, E, F, H, I, K, L, N and O) micrographs. A–C, *Quercus* section *Cyclobalanopsis* sp. 2; D–F, *Quercus* section *Cyclobalanopsis* sp. 3; G–I, *Quercus* section *Ilex* sp. 1; J–L, *Quercus* section *Ilex* sp. 2; M–O, *Quercus* section *Ilex* sp. 3. Scale bars: 10 µm (A, B, D, E, G, H, J, K, M and N); 1 µm (C, F, I, L and O).

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2014). The *Quercus* section *Ilex* sp. 1 pollen type, and the following section *Ilex* pollen types described here, share many of the morphological features documented for the previously mentioned fossil pollen records.

Quercus section *Ilex* sp. 2 (Fig. 3J–L)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 22–23 µm long in LM, 21–22 µm in SEM, equatorial diameter 19–20 µm in LM, 17–18 µm in SEM; tricolporoidate, colpi long, pori small and indistinct; exine 0.6–0.7 µm thick (LM); tectate; sculpture psilate in LM, nanorugulate and perforate in SEM, nanorugulae varying in size and shape, irregularly distributed, densely packed, forming desert-rose-like clusters/agglomerates (SEM).

Remarks: Grains assigned to *Quercus* section *Ilex* sp. 2 differ from the previous section *Ilex* sp. 1 pollen type by the shorter and narrower nanorugulae and the typical desert-rose-like sculpture formed by the agglomerated nanorugulae. Such sculpture arrangement is typical for pollen from several extant species of *Quercus* section *Ilex*, e.g. *Quercus baloot* Griff., *Quercus coccifera* L., *Quercus engleriana* Seemen, *Quercus ilex* L. and *Quercus lanata* Sm. (Denk & Tekleva, 2014). Again, it is a pollen type occurring in various extant Mediterranean and Asian species of oaks (De Langhe, 2022).

Quercus section *Ilex* sp. 3 (Fig. 3M–O)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 26–27 µm long in LM, 22–23 µm in SEM, equatorial diameter 21–22 µm in LM, 19–20 µm in SEM; tricolporoidate, colpi long, pori small and indistinct; exine 0.7–0.8 µm thick (LM); tectate; sculpture scabrate in LM, nanorugulate (rod-shaped), fossulate and perforate in SEM, nanorugulae indistinct, irregularly arranged, clustered/agglomerating (SEM).

Remarks: The *Quercus* section *Ilex* sp. 3 to sp. 5 pollen types are all similar and overlap considerably regarding size of the grains, the pollen wall thickness, aperture arrangements and the sculpture. As observed with SEM the basal sculpture elements in all three pollen types are nano- to micro-sized rugulae (rod-shaped) as well as fossulae and perforations. Sculpture wise, the section *Ilex* sp. 3 pollen type has the most indistinct rugulae that are covered/masked by sporopollenin. This is less apparent in the section *Ilex* sp. 5 pollen type where the rugulae are more distinct, and in the section *Ilex* sp. 4 pollen type the rugulae are not covered and clearly portrayed. It is possible that this difference in the sculpture is reflecting actual species differences, but it may also be that it merely documents the variability within a single biological taxon. Similar variability has been documented for several extant Mediterranean and Asian species in *Quercus* section *Ilex*, for example *Q. coccifera, Quercus dolicholepis* A.Camus, *Quercus franchetii* Skan, *Q. ilex* and *Quercus semecarpifolia* Sm. by Denk & Tekleva (2014).

Quercus section *Ilex* sp. 4 (Fig. 4A–C)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 28–29 µm long in LM, 24–25 µm in SEM, equatorial diameter 15–16 µm in LM, 12–13 µm in SEM; tricolporoidate, colpi long, pori small and indistinct; exine 0.8–0.9 µm thick (LM); tectate; sculpture scabrate in LM, nano- to microrugulate (rod-shaped), fossulate and perforate in SEM, rugulae distinct, 0.3–0.8 µm long, 0.09–0.12 µm wide, densely packed, irregularly arranged, clustered/agglomerating, often fused/branched (SEM).

Quercus section *Ilex* sp. 5 (Fig. 4D–F)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 26–27 µm long in LM, 23–24 µm in SEM, equatorial diameter 23–24 µm in LM, 21–22 µm in SEM; tricolporoidate, colpi long, pori small and indistinct; exine 0.9–1.0 µm thick (LM); tectate; sculpture psilate in LM, nano- to microrugulate, fossulate and perforate in SEM, rugulae indistinct 0.3–0.8 µm long, 0.08–0.15 µm wide, densely packed, irregularly arranged, clustered/agglomerating (SEM).

Subgenus: Quercus

Section: Quercus

Quercus section *Quercus* sp. (Fig. 4G–I)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 27–28 µm long in LM, 27–28 µm in SEM, equatorial diameter 24–25 µm in LM, 23–24 µm in SEM; tricolporoidate, colpi long, pori small and indistinct; exine 1.2–1.3 µm thick (LM); tectate; sculpture scabrate in LM, granulate to areolate, fossulate and perforate in SEM.

Remarks: The LM- and SEM-based pollen morphology of *Quercus* section *Quercus* has been documented in detail by Crepet & Daghlian (1980); Miyoshi (1982); Solomon (1983b); Wang & Pu (2004); Denk & Grimm (2009); Makino *et al.* (2009); Li *et al.* (2011) and Miyoshi *et al.*

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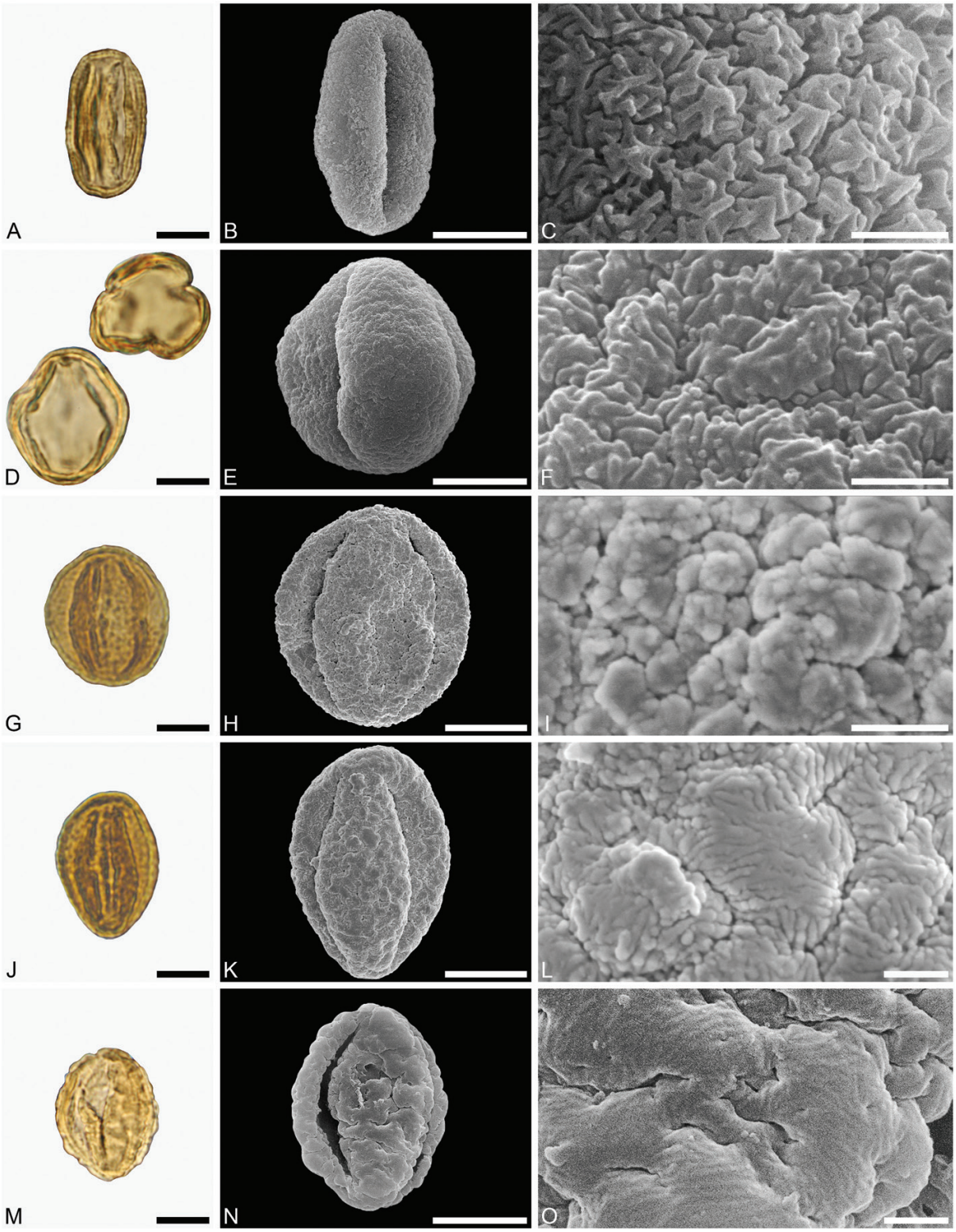

Figure 4. Fossil Fagaceae pollen from the Ban Pa Kha Subbasin. Light microscope (A, D, G, J and M) and scanning electron microscope (B, C, E, F, H, I, K, L, N and O) micrographs. A–C, *Quercus* section *Ilex* sp. 4; D–F, *Quercus* section *Ilex* sp. 5; G–I, *Quercus* section *Quercus* sp.; J–L, Fagaceae gen. & spec. indet. 1; M–O, Fagaceae gen. & spec. indet. 2. Scale bars: 10 µm (A, B, D, E, G, H, J, K, M and N) and 1 µm (C, F, I, L and O).

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(2011). The ultrastructure (TEM) of section *Quercus* has been studied to a lesser extent, but see Wang & Pu (2004) and Denk & Tekleva (2014). The areolate sculpture elements, observed in the fossil *Quercus* section *Quercus* pollen type, that are apparently composed of smaller agglomerated elements but masked/covered by sporopollenin is a typical feature for this section (see fig. 3 in Denk & Grimm, 2009). The palynological record of *Quercus* section *Quercus* type pollen, documented using combined LM and SEM, includes dispersed grains from the Mid-Eocene (Late Lutetian to Early Bartonian) of Hareøen, western Greenland (Grímsson *et al.*, 2015), the Late Eocene (Priabonian) of Florissant, Colorado, USA (Bouchal *et al.*, 2014), the Late Oligocene of south-western Siberia (Denk *et al.*, 2021a), the Miocene of eastern China (Liu *et al.*, 2007), the Mid-Miocene of Anatolia, Turkey (Bouchal *et al.*, 2016a, 2017; Bouchal, 2019), the Late Miocene of Iceland (Denk *et al.*, 2010) and north-western Greece (Bouchal *et al.*, 2020) and the Pleistocene of south-eastern Albania (Denk *et al.*, 2021b). Some of these records bear similarities to the fossil *Quercus* section *Quercus* sp. pollen type presented here.

Subfamily: not determined

Genus: not determined

Fagaceae gen. & spec. indet. 1 (extinct) (Fig. 4J–L)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 28–29 µm long in LM, 27–28 µm in SEM, equatorial diameter 19–20 µm in LM, 19–20 µm in SEM; tricolporoidate, colpi long, pori indistinct; exine 1.0–1.2 µm thick (LM); tectate; sculpture scabrate in LM, rugulate to areolate, fossulate and perforate in SEM, rugulae/areolae with microrugulate to striate suprasculpture, microrugulae/striae parallel arranged in groups (SEM).

Remarks: The Fagaceae gen. et spec. indet. 1 type pollen is quite unique, and we are unaware of any living Fagaceae/*Quercus* producing pollen or any fossil pollen showing the same combination of features as depicted here. The aperture arrangement, especially the indistinct pori, in combination with the sculpture observed with SEM suggest that the pollen should be affiliated with Fagaceae. It is possible that this pollen type represents some extinct fagaceous taxon or it could be an aberrant pollen from one of the extant sections already mentioned.

Fagaceae gen. & spec. indet. 2 (extinct) (Fig. 4M–O)

Description: Pollen, monad, isopolar, prolate, trilobate in polar view, elliptic in equatorial view; polar axis 25–26 µm long in LM, 22–23 µm in SEM, equatorial diameter 19–20 µm in LM, 16–17 µm in SEM; tricolporoidate, colpi long, pori indistinct; exine 1.1–1.3 µm thick (LM); tectate; sculpture scabrate in LM, areolate, fossulate and perforate in SEM, areolae with indistinct rugulate to striate suprasculpture, rugulae/striae parallel arranged in groups (SEM).

Remarks: The Fagaceae gen. et spec. indet. 2 pollen type is similar to the previous indet. type, but it is smaller and has fewer and larger areolae and the rugulate/striate suprasculpture is less distinct.

DISCUSSION

Early Miocene Fagaceae of northern Thailand in the context of Cenozoic diversity of Fagaceae in Southeast and East Asia

This is the first detailed study into Fagaceae diversity during the Late Palaeogene/Early Neogene of Thailand. We recognized 15 fossil species of Fagaceae belonging to two subfamilies in the pollen assemblage of the Li Basin (Table 2). Four species belong to extinct fagaceous lineages (*Eotrigonobalanus*, two species; Fagaceae gen. & spec. indet., two species). The remaining species can be assigned to three sections in two subgenera of *Quercus* (sections *Cyclobalanopsis*, *Ilex* and *Quercus*), and to unresolved taxa of subfamily Castaneoideae, of which one pollen morphotype shows morphological affinities with the extant genera *Castanopsis* and *Lithocarpus*. Overall, this suggests that except for the extinct taxa, most of the fagaceous diversity in the Late Palaeogene/Early Neogene of northern Thailand is also found in the modern vegetation of Thailand.

A strikingly similar taxonomic composition and diversity of Fagaceae has previously been reported from the Mid-Eocene strata of southern China (Changchang Fm, Hainan Island; Hofmann, 2010; Spicer *et al.*, 2014), including the extinct *Eotrigonobalanus* and sections *Cyclobalanopsis*, *Ilex* and *Quercus* among oaks. In addition, several Castaneoideae pollen morphotypes were recorded from the Changchang Fm. The leaf fossil record from the Changchang Fm (Spicer *et al.*, 2014; Liu *et al.*, 2020) included *Castanopsis*, *Lithocarpus*, extinct Castaneoideae and *Quercus* section *Cyclobalanopsis*. Palaeoclimate estimates derived from leaf physiognomy suggested a fully humid cool tropical climate for the Changchang leaf assemblage, whereas the palynological record with diverse oaks of section *Ilex* points to precipitation seasonality and a possible monsoon influence. Pollen assemblages similar to the one from the Li Basin are further known from the Late Eocene/Early Oligocene Maoming Basin, South China (Herman *et al.*, 2017). The pollen flora from the Maoming Basin records Fagaceae, Dipterocarpaceae

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Table 2. Taxonomy of dispersed Fagaceous pollen from the Ban Pa Kha Subbasin

Fagaceae taxon	Stratigraphy of the Ban Pa Kha Subbasin (Li Basin)		
	Interburden	Upper coal seam	Overburden
Eotrigonobalanus sp.1	X	X	X
Eotrigonobalanus sp.2		X	X
Castaneoideae gen. et spec. indet. 1	X	X	
Castaneoideae gen. et spec. indet. 2	X	X	X
Quercus sect. *Cyclobalanopsis* sp. 1	X		
Quercus sect. *Cyclobalanopsis* sp. 2	X		
Quercus sect. *Cyclobalanopsis* sp. 3			X
Quercus sect. *Ilex* sp. 1	X	X	X
Quercus sect. *Ilex* sp. 2	X		X
Quercus sect. *Ilex* sp. 3	X	X	X
Quercus sect. *Ilex* sp. 4	X		X
Quercus sect. *Ilex* sp. 5			X
Quercus sect. *Quercus* sp.	X		
Fagaceae gen. et spec. indet.1	X		
Fagaceae gen. et spec. indet.2	X		X

and Pinaceae as dominant elements. Among Fagaceae, *Eotrigonobalanus*, unspecified *Quercus*, *Quercus* spp. (verrucate; possibly corresponding to *Quercus* section *Quercus*), *Quercus* section *Cyclobalanopsis* and *Castanopsis* were identified. The Early Oligocene leaf assemblage from the Maoming Basin suggested a clear monsoon signal for this flora (Herman *et al.*, 2017). From roughly coeval strata in the Wenshan Basin, south-western China, Li *et al.* (2015b; see Tian *et al.*, 2021 for a corrected age) reported an Early Oligocene pollen assemblage with *Castanopsis*, Castaneoideae (as *Castanea*), *Quercus* unspecified and *Quercus* section *Cyclobalanopsis*. Overall, the spore and pollen assemblage from Wenshan suggested a seasonal climate with weak monsoon intensity.

Well-preserved wood remains from the Oligocene Yongning Fm, South China, comprised the Fagaceae *Lithocarpus* (one species) and *Castanopsis* (two species; Huang *et al.*, 2018). The preserved wood characteristics, weak to absent growth ring boundaries and prominent ring-porous wood, suggested a tropical monsoon climate for this part of China during the Oligocene. Finally, Mu *et al.* (2015) described fruits of *Lithocarpus* from Miocene strata of the Nanlin Fm (Yunnan, south-western China). The age for this formation is not well constrained, but on the basis of palynology, an Early to Mid-Miocene age has been suggested.

Present vegetation in Thailand and climatic and biotic envelopes of forest trees in Fagaceae

Forest vegetation in the western and north-western parts of Thailand comprises remnants of closed monsoon forests in the lowlands, dry dipterocarp savannah forests at lower elevations and montane pine-oak forests at mid to high elevations (Menitsky, 2005). Closed monsoon forests contain a rare *Quercus* section *Cyclobalanopsis*, whereas higher up, dipterocarp forests occur on dry soils and mixed dipterocarp-Fagaceae forests on mesic soils of gentle slopes. The latter forests are rich in Fagaceae (*Quercus*, *Lithocarpus* and *Castanopsis*). On dry soils, *Quercus* section *Ilex* thrives in the upper tropical belt.

Above *c*. 900 m, dipterocarp forest is replaced by subtropical dry or wet pine-oak or oak (dominated by Fagaceae) forest. The relatively drier pine-oak forest is dominated by diverse oaks of all sections of subgenus *Cerris* and by *Lithocarpus* spp. and *Castanopsis* spp. In some places, species of *Quercus* section *Ilex* grow up to 2000 m a.s.l.

The more mesic 'oak' forests on moist soils are closed evergreen forests with the dominance of Fagaceae (*Lithocarpus*, *Castanopsis* and *Quercus* sections *Ilex* and *Cyclobalanopsis*). These latter forests contain typical Cenozoic relict taxa such as *Diospyros* L., *Engelhardia* Lesch. ex Blume, *Euodia* J.R.Forst. & G.Forst. *Manglietia* Blume (=*Magnolia* L.), *Styrax* L. and others. In addition, temperate taxa such as *Betula* L., *Carpinus* L. and *Rhododendron* L. form part of these forests (Menitsky, 2005; Santisuk, 2012).

According to the Köppen–Geiger climate classification, Thailand is characterized by a 'tropical savannah' climate (*Aw*, Kottek *et al.*, 2006; Peel, Finlayson & McMahon, 2007). This is in accordance with the grassy ground cover found in many pine-oak forests and corresponds to the Tropical and Subtropical Dry Broadleaf Forests biome of Olson *et al.* (2001) and

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its wetter variant, the Tropical and Subtropical Moist Broadleaf Forests biome. Overall, the fossil record of Fagaceae from the Li Basin suggests that similar forests existed in northern Thailand since the start of the Early Neogene, if not since the Palaeogene (see the previous section).

Origin of seasonal tropical Fagaceae-dominated forests in northern Thailand

Fossil assemblages with a particularly high diversity of Fagaceae have been reported from several mid- to high-latitude localities in the Northern Hemisphere (e.g. Denk *et al.*, 2012, Central Europe; Bouchal *et al.*, 2014, western North America; Pavlyutkin, Chekryzhov & Petrenko, 2014, Russian Far East; Grímsson *et al.*, 2016a, western Greenland; Grímsson *et al.*, 2016b, Central Europe; Prader *et al.*, 2020, eastern North America; Sadowski *et al.*, 2020, Baltic amber region of North Europe). In these areas, hyper-diverse fagaceous assemblages were of ephemeral nature coinciding with warm phases in Earth history (western North America, Greenland, Baltic amber, Far East) and their diversity decreased dramatically during the Neogene when the modern vegetation in these areas developed (e.g. Mai, 1995; Graham, 1999).

In contrast, characteristic forest communities of tropical lowlands and mid-elevations (Tropical and Subtropical Moist Broadleaf Forests, Tropical and Subtropical Dry Broadleaf Forests) in Southeast Asia might have existed since the Eocene, reflected by fossil plant assemblages (pollen and leaf assemblages), with the dominance of Fagaceae and variably co-dominating Dipterocarpaceae or Pinaceae (e.g. Herman *et al.*, 2017). The presence of species of *Quercus* sections *Cyclobalanopsis* and *Ilex* is important in distinguishing wet forests from monsoonal, seasonal forest vegetation. For example, Huang *et al.* (2022) reported forest vegetation with Fagaceae and Dipterocarpaceae from the Oligocene Ha Long flora of Vietnam. Extinct Castaneoideae (*Berryophyllum* Jones & Dilcher and *Castaneophyllum* Jones & Dilcher), *Lithocarpus* and *Quercus* section *Cyclobalanopsis* represent a lowland tropical evergreen forest corresponding to a modern vegetation in southern Thailand and less so to the palaeovegetation of the Ban Pa Kha Subbasin of the Li Basin. However, extant members of *Quercus* sections *Cyclobalanopsis* and *Ilex* can also be found in northern Thailand (Phengklai, 2006; Santisuk, 2012).

Our recognition of tropical montane pine-oak and mesic 'oak' (Fagaceae-dominated) forest in the Late Oligocene to Early Miocene Ban Pa Kha Subbasin of the Li Basin, based on the high diversity and proportion of fossil Fagaceae pollen (Tables 1 and 2), conflicts with previous interpretations of the palynological record from Cenozoic deposits of Thailand (Morley *et al.*, 2001; Songtham *et al.*, 2005), which suggested a north temperate (Late Oligocene) flora that would have been replaced by a tropical flora after the Early Miocene. According to these authors, sediment fill of the Ban Pa Kha Subbasin would have coincided with temperate conditions during the Late Oligocene/Early Miocene. We interpret the varying contributions of 'temperate' and 'tropical' taxa in the dispersed spore and pollen record of Thailand as reflecting different facies and lowland/upland settings. This is in agreement with previous findings of macrofossils in the Ban Pa Kha Subbasin that could be affiliated with subtropical/tropical taxa (Grote & Srisuk, 2021).

A typical temperate element in the Late Oligocene/Early Miocene flora of Thailand would be *Fagus* (Table 3). However, previous leaf records (Endo, 1967) and pollen records (Songtham *et al.*, 2003; Sattraburut *et al.*, 2021) clearly indicate alliances outside Fagaceae. Endo (1967) assigned leaf remains from the Ban Pa Kha Subbasin as *Fagus feroniae* Unger, a taxon that has subsequently been transferred to *Alnus* Mill. (Kvaček & Holý, 1974). Similar leaf morphotypes are not only encountered in modern *Alnus*, but also in Dipterocarpaceae (see Khan *et al.*, 2020). Pollen described as *Faguspollenites* by Songtham *et al.* (2003) only superficially resembles *Fagus* but differs by its wide colpi and sculpture elements of the colpus membrane that are much too large for *Fagus*. Likewise, a pollen grain figured by Sattraburut *et al.* (2021) from Miocene strata of Laos close to the Li Basin and referred to as *Fagus*, clearly differs from this genus by the structure of its pori and the conspicuous thickening in the colpus-porus region (suggesting Cornaceae/*Nyssa* Gronov. ex L.). Therefore, records of *Fagus* from the Late Oligocene and Miocene of Thailand and adjacent regions would need to be re-investigated. In combination with a highly diverse assemblage of *Quercus* pollen types representing the evergreen sections *Cyclobalanopsis* and *Ilex*, along with diverse Castaneoideae and tropical elements among leaf macrofossils (cf. Sawangchote, 2003; Sawangchote *et al.*, 2009, 2010) and fossil pollen (Table 3), we note the following: (1) The clear distinction between temperate and tropical forest associations is not given whereby the Late Oligocene/Early Miocene assemblages have a distinct tropical component; (2) The presence of evergreen oak lineages indicative of seasonal climates would suggest that the Tropical and Subtropical Moist/Dry Forests biomes were in place by the Late Palaeogene; and (3) The change to true tropical forest assemblages with increased numbers of tropical taxa, such as Dipterocarpaceae, might as well reflect changes from lowland to upland vegetation, slope aspect and soil moisture. Hence, in addition to reflecting a profound climatic change across the Palaeogene/Neogene boundary, these different plant associations may also reflect different facies/

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Table 3. Palaeobotanical records from the Ban Pa Kha Subbasin, Li Basin. Abbreviations: L, leaf; F, fruit; P, sporomorph; H, herb; S, shrub; T, tree; W, wetland; B, broadleaf forest; D, deciduous forest; E, evergreen forest; C, (upland) conifer forest; V, various habitats

Fossil taxon	Organ	Reference	Ecology		Remark
Ferns					
Davilliaceae					
Polypodiisporites minimus	P	Songtham *et al.* (2005)	H	V	
Osmundaceae					
Osmunda sp.	P	Malaikanok (2021)	H	V	
Polypodiaceae					
Laevigatosporites haardtii	P	Songtham *et al.* (2005)	H	V	
Polypodiisporites inangahuensis	P	Songtham *et al.* (2005)	H	V	
Polypodiisporites radiatus	P	Songtham *et al.* (2005)	H	V	
Pteridaceae					
Striatriletes susannae	P	Songtham *et al.* (2005)	H	V	
Polypodiisporites retirugatus (*Pteris*)	P	Songtham *et al.* (2005)	H	V	
Gymnosperms					
Araucariaceae					
Araucaria sp.	P	Songtham *et al.* (2005)	T	E	
Cupressaceae					
Glyptostrobus europaeu / *Glyptostrobus* sp.	L	Endo (1967)	T	W	
		Grote (2015)			
Inaperturopollenites sp. (*Taxodium*/*Sequoia*)	P	Songtham *et al.* (2005)	T	V	
Sequoia langsdorfii	L	Endo (1963, 1967)	T	E, B, C	
Taiwania sp.	L	Grote (2015)	T	E, B, C	
Taxodium thaiensis/ *Taxodiaceaepollenites* sp.	L, P	Endo (1963)	T	W, D	
		Songtham *et al.* (2003, 2005)	T		
Pinaceae					
Piceaepollenites sp. (*Picea*)	P	Songtham *et al.* (2003, 2005)	T	C	
Pinus sp./*Pinuspollenites* sp.	F, P	Grote & Srisuk (2021)	T	W, B, E	
		Songtham *et al.* (2003, 2005)			
		Malaikanok (2021)			
Tsugaepollenites igniculus / *Tsuga* spp.	P	Songtham *et al.* (2003, 2005)	T	C	
		Malaikanok (2021)			
Podocarpaceae					
Podocarpidites ellipticus (*Podocarpus*)	P	Songtham *et al.* (2005)	T	E, C	

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Table 3. Continued

Fossil taxon	Organ	Reference	Ecology		Remark
Sciadopityaceae					
Sciadopitys spp.	L, P	Grote (2015) Malaikanok (2021)	T	B, C	
Angiosperms					
Altingiaceae					
Liquidambarpollenites stigmosus (*Liquidambar*)	P	Songtham *et al.* (2003, 2005), Malaikanok (2021)	T	W, D	
Anacardiaceae					
Mangifera paleoindica	L	Sawangchote *et al.* (2009)	T	E	
Mangifera paleocaloneura	L	Sawangchote *et al.* (2009)	T	E	
Semecarpus paleocurtisiodes	L	Sawangchote *et al.* (2010)	T	D	
Aquifoliaceae					
Ilexpollenites iliacus (*Ilex*)	P	Songtham *et al.* (2003, 2005), Malaikanok (2021)	T	W, D, E	
Araliaceae					
Araliaceae gen. et spec. indet.	P	Malaikanok (2021)	T, S	B, E	*Aralia*/*Polyscias*/ *Schefflera*
Betulaceae					
Alnus thaiensis	L	Endo (1963, 1967)	T	W	
Alnipollenites verus (*Alnus*)	P	Songtham *et al.* (2003, 2005), Malaikanok (2021)	T	W	
Betula spp.	P	Songtham *et al.* (2003, 2005), Malaikanok (2021)	T	D	
Carpinus sp.	L	Endo (1963)	T	D	
Momipites coryloides (*Corylus*)	P	Songtham *et al.* (2003, 2005), Malaikanok (2021)	T	D	
Caprifoliaceae					
Loniceraepollenites sp. (*Lonicera*)	P	Songtham *et al.* (2003, 2005)	S	D	
Cyperaceae					
Sparganium thaiensis	L	Endo (1963)	H	W	
Dipterocarpaceae					
Dipterocarpaceae gen. et spec. indet	P	Unpublished data	T	D, E	
Ebenaceae					
Diospyros sp.	P	Malaikanok (2021)	T	W, D, E	
Fabaceae					
Albizia paleoproceroides	L	Sawangchote (2003)	T	D, E	
Antheroporum paleolamphunensis	L	Sawangchote (2003)	T	D, E	
Fagaceae					
Fagus feroniae	L	Endo (1967)	T	B, D, E	cf. *Alnus*
Faguspollenites sp. (*Fagus*)	P	Songtham *et al.* (2003, 2005)	T	B, D, E	Not *Fagus*

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Table 3. Continued

Fossil taxon	Organ	Reference	Ecology		Remark
Quercus lanceaefolia	L	Endo (1967)	T	B, D, E	*Q*. sect. *Cyclobalanopsis*
Quercus protoglauca	L	Endo (1967)	T	B, D, E	*Q*. sect. *Cyclobalanopsis*
Quercus/*Castanopsis* sp.	L	Grote (2005)	T	B, D, E	*Q*. sect. *Cyclobalanopsis*
Quercoidites sp. (*Quercus*)	P	Songtham *et al.* (2003, 2005)	T	B, D, E	
Hamamelidaceae					
Hamamelidaceae gen. et spec. indet.	P	Malaikanok (2021)	T, S	W, B, E, D	cf. *Parrotia*
Juglandaceae					
Caryapollenites simplex (*Carya*)	P	Songtham *et al.* (2003, 2005)	T	W, D	
Juglanspollenites versus (*Juglans*)	P	Songtham *et al.* (2003, 2005)	T	W, D	
Juglandinae spp. *(Juglans/Pterocarya)*	P	Malaikanok (2021)	T	W, D	
Pterocaryapollenites stellatus (*Pterocarya*)	P	Songtham *et al.* (2003, 2005)	T	W, D	
Lythraceae					
Lagerstroemia sp.	P	Songtham *et al.* (2005)	T, S	D, E	
Moraceae					
Ficus eowightiana	L	Endo (1967)	T, S, C	D, E	
Oleaceae					
Ligustrum sp.	P	Malaikanok (2021)	S, T	W, B, E	cf. Hamamelidaceae
Retitrescolpites (aff. Oleaceae)	P	Songtham *et al.* (2005)	T, S	V	cf. *Viburnum*
Pentaphylacaceae					
Adinandra sp.	P	Malaikanok (2021)	T, S	W, B, D, E	
Poaceae					
Poaceae gen. et spec. indet.	P	Malaikanok (2021)	H	V	
Salicaceae					
Salix sp.	P	Malaikanok (2021)	T, S	V	
Sapotaceae					
Sapotaceae spp.	P	Malaikanok (2021)	T, S	V	
Styracaceae					
Rehderodendron sp.	P	Malaikanok (2021)	T	B, E	
Styrax sp.	P	Malaikanok (2021)	T, S	B, E	
Symplocaceae					
Symplocos sp.	P	Malaikanok (2021)	S, T	E	
Trapaceae					
Sporotrapoidites medius (aff. *Trapa*)	P	Songtham *et al.* (2005)	H	W	

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Table 3. Continued

Fossil taxon	Organ	Reference	Ecology		Remark
Ulmaceae					
Ulmipollenites (aff. *Ulmus*)	P	Songtham *et al.* (2003, 2005), Malaikanok (2021)	T, S	D	

environmental conditions. Sepulchre *et al.* (2010) reported high amounts of Poaceae pollen in the Mid-Miocene deposits of Thailand. Poaceae are co-dominant with *Castanopsis*, *Quercus* and *Pinus*, reflecting a dry tropical pine-oak forest with grass ground cover (Santisuk, 2012).

In summary, our new data suggest that seasonal (monsoon) tropical climates, which were probably cooler than they are today, and their corresponding forest vegetation existed in northern Thailand at least since the Late Palaeogene and persisted into the modern vegetation of Thailand. Apparently, no sharp change from (northern) temperate to tropical environments occurred between the Late Oligocene and the Mid-Miocene.

Although *Quercus* sections *Cyclobalanopsis* and *Ilex* are abundant in the Late Oligocene/Early Miocene flora of the Li Basin, deciduous or semi-deciduous elements of *Quercus* section *Cerris* are absent from the fossil pollen assemblage. In contrast, the latter is part of contemporaneous dry oak-pine forests (Menitsky, 2005). Likewise, *Quercus* section *Cerris* is absent from the Palaeogene forest in Southeast Asia and the southern Himalayas (Hofmann, 2010; Denk & Bouchal, 2021). Members of *Quercus* section *Cerris* are essentially temperate species (Naryshkina & Evstigneeva, 2020; Denk & Bouchal 2021) and may have reached tropical areas during a secondary radiation in the later part of the Miocene.

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DATA AVAILABILITY

The data underlying this article are available in the article.

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